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$$\frac{m_1(u_1+v_1)(u_1-v_1)}{m_1(u_1-v_1)}=\frac{m_2(v_2+u_2)(v_2-u_2)}{m_2(v_2-u_2)}$$

$$u_1+v_1=v_2+u_2$$

$$u_1-u_2=v_2-v_1 \quad \text{Rearranging,} \quad (4.50)$$

Equation (4.50) can be rewritten as

$$u_1-u_2=-(v_1-v_2)$$

This means that for any elastic head on collision, the relative speed of the two elastic bodies after the collision has the same magnitude as before collision but in opposite direction. Further note that this result is independent of mass.

Rewriting the above equation for v_1 and v_2,

$$v_1=v_2+u_2-u_1 \quad (4.51)$$

Or

$$v_2=u_1+v_1-u_2 \quad (4.52)$$

To find the final velocities v_1 and v_2:
Substituting equation (4.52) in equation (4.47) gives the velocity of m_1 as

$$m_1(u_1-v_1)=m_2(u_1+v_1-u_2-u_2)$$

$$m_1(u_1-v_1)=m_2(u_1+v_1-2u_2)$$

$$m_1u_1-m_1v_1=m_2u_1+m_2v_1-2m_2u_2$$

$$m_1u_1-m_2u_1+2m_2u_2=m_1v_1+m_2v_1$$

$$(m_1-m_2)u_1+2m_2u_2=(m_1+m_2)v_1$$

$$\text{or } v_1=\left(\frac{m_1-m_2}{m_1+m_2}\right)u_1+\left(\frac{2m_2}{m_1+m_2}\right)u_2 \quad (4.53)$$

Similarly, by substituting (4.51) in equation (4.47) or substituting equation (4.53) in equation (4.52), we get the final velocity of m_2 as

$$v_2=\left(\frac{2m_1}{m_1+m_2}\right)u_1+\left(\frac{m_2-m_1}{m_1+m_2}\right)u_2 \quad (4.54)$$

Case 1: When bodies has the same mass i.e., $m_1 = m_2$,

$$\text{equation (4.53)} \Rightarrow v_1=(0)\,u_1+\left(\frac{2m_2}{2m_2}\right)u_2$$

$$v_1=u_2 \quad (4.55)$$

$$\text{equation (4.54)} \Rightarrow v_2=\left(\frac{2m_1}{2m_1}\right)u_1+(0)u_2$$

$$v_2=u_1 \quad (4.56)$$

The equations (4.55) and (4.56) show that in one dimensional elastic collision, when two bodies of equal mass collide after the collision their velocities are exchanged.

Case 2: When bodies have the same mass i.e., $m_1 = m_2$ and second body (usually called target) is at rest ($u_2 = 0$),

By substituting $m_1 = m_2$ and $u_2 = 0$ in equations (4.53) and equations (4.54) we get,

$$\text{from equation (4.53)} \Rightarrow v_1=0 \quad (4.57)$$

$$\text{from equation (4.54)} \Rightarrow v_2=u_1 \quad (4.58)$$

Equations (4.57) and (4.58) show that when the first body comes to rest the second